ASSIGNMENT 9

1. **The major limitation of a Gantt chart is:**
   A. **It does not show activity precedence or time variability**
   B. It cannot display activities across time periods
   C. It shows activities but not their durations
   D. It requires too many resources to prepare and update
2. **The techniques used as network-modelling in project management are:**
   A. Critical Path Method and PERT
   B. Lean Manufacturing and Six Sigma
   C. Gantt Chart and Work Breakdown Structure
   D. Brainstorming and SWOT analysis.
3. **Advantages of network modelling in project management is**
   A. Increases project complexity and delays timeline
   B. Focuses only on financial aspects ignoring scheduling
   C. Eliminates the need for resource planning entirely
   D. **Shows activity relationships, reduces project duration and cost**
4. **The type of project where the CPM technique mainly applied is:**
   A. R&D projects with uncertain and variable timelines
   B. **Small projects with well-defined activity durations**
   C. Projects aimed only at reducing environmental impact
   D. Large government projects with many stakeholders
5. **The 'critical path' in a project network is determined as:**
   A. **The path with the longest project completion time**
   B. The route that starts first in the project timeline
   C. The path with the least number of activities
   D. The path that uses the fewest project resources
6. **The role of 'slack time' in network analysis is:**
   A. To define the earliest start time of each activity
   B. To show how long an activity can be delayed
   C. To indicate total time spent on critical activities

D. To represent unused resources in the project

7. **The symbol in network diagrams used to indicate the start or end of an activity is:**

   A. A rectangular box

   B. **A circular node**

   C. A solid arrow

   D. A dotted line

8. **The critical path in a PERT network is defined as:**

   A. The shortest path connecting starts to finish nodes

   B. Activities on the path with zero slack time

   C. The sequence with the maximum total slack time allowed

   D. Activities with the highest resource requirements

9. **The variance of an activity's completion time in PERT is calculated as:**

   A. $\frac{t_o + t_p}{2}$

   B. $\left(\frac{t_p - t_o}{6}\right)^2$

   C. $\sqrt{\left(\frac{t_p - t_m}{3}\right)}$

   D. $\left(\frac{t_m - t_o}{3}\right)$

   Where:
   (t_o) = Optimistic time
   (t_m) = Most likely time
   (t_p) = Pessimistic time

10. **When project activities are crashed to shorten duration, costs usually:**

   A. First decreases than increases

   B. Decreases

   C. Increases

   D. Remain unaffected